

Instructions - Transport Module

1. Overview

This system implements a Transport Management Module for a supermarket chain.

The system supports:

- Managing trucks, drivers, and sites
- Creating deliveries
- Validating driver-truck compatibility
- Checking weight constraints
- Managing delivery documents and transported items

2. How to Run the System

Option 1 - Using JAR file (Recommended)

1. Open terminal in the project root folder
2. Run:

```
java -jar release/adss2025_v01.jar
```

Option 2 - Running from source code

1. Navigate to dev folder
2. Compile:

```
javac Main.java domain\*.java enums\*.java repository\*.java
```
3. Run:

```
java Main
```

3. System Initialization (Sample Data)

The system initializes the following data:

Delivery Zone

- South Zone

Sites

- Supplier: "Tnuva Supplier"
- Branch: "Beer Sheva Branch"
- Branch: "Dimona Branch"

Truck

- License: 123-45-678
- Model: Volvo FH
- Max Capacity: 15000

Drivers

- Valid driver (License C)
- Invalid driver (License B)

Items

- Milk (100 units)
- Bread (50 units)
- Cheese (30 units)

4. System Behavior

Valid Delivery

- Driver license matches truck
- Weight is within allowed range
 - Delivery is created successfully
 - Status is set to READY

Invalid Delivery

- Driver license does NOT match truck
 - Delivery creation fails

Overweight Scenario

- If weight exceeds maximum capacity
 - Delivery is rejected
 - Status is set to OVERWEIGHT

5. Output Description

The system prints:

- Delivery details (ID, date, time)
- Source site
- Driver and truck information
- Weight and status

- List of destinations
- Items per destination

6. Notes

- All data is stored in memory (no database)
- The system includes validation logic for:
 - Driver-truck compatibility
 - Weight constraints
 - Valid delivery documents

7. Unit Tests

The unit tests can also be executed from the project root folder.

Compile the files

```
javac -cp ".;dev\lib\junit-platform-console-standalone-1.10.2.jar" dev\Main.java  
dev\domain\*.java dev\enums\*.java dev\repository\*.java dev\test\*.java
```

Run the tests

```
java -jar dev\lib\junit-platform-console-standalone-1.10.2.jar --class-path dev --scan-class-path
```

Notes

- All data is stored in memory only.
- The system includes sample data initialized in Main.
- The program demonstrates both a valid delivery creation and an invalid delivery scenario.
- The tests validate core transport logic such as driver–truck compatibility, weight validation, and delivery creation.